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Norm inequalities for certain classes of functions and their Fourier transforms

Barry Booton, Yoram Sagher *

Department of Mathematical Sciences, Florida Atlantic University, 777 Glades Road, Boca Raton, FL 33431, USA

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Abstract

We apply tools of interpolation theory and a commutative property of the Hilbert transform to prove necessary and sufficient conditions related to trigonometric series. These results extend and improve related theorems proven by several authors, summarized by Boas. In addition, we explore inequalities and operators, both connected to Hardy's inequalities, on certain classes of functions, including quasimonotone functions.
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1. Introduction

For a sequence $\{\lambda_n\}$, we denote by λ_n^* the nonincreasing rearrangement of $|\lambda_n|$. Hardy and Littlewood [5] showed that for $q \geqslant 2$, a necessary and sufficient condition that λ_n be, for every rearrangement, the Fourier coefficients of a function $f \in L^q(0,\pi)$ is that $\sum(\lambda_n^*)^q n^{q-2} < \infty$; and then

$$\|f\|_{L^q(0,\pi)} \leqslant C\left(\sum\left(\lambda_n^*\right)^q n^{q-2}\right)^{\frac{1}{q}} \tag{1}$$

for every such f. (Throughout we denote by C a generic constant that does not depend on the functions in the inequalities.) They also showed that if $f(x) = \sum \lambda_n \cos nx \in L^q$, $1 < q \leqslant 2$,

* Corresponding author. Fax: +1 (561) 297 2436.
E-mail addresses: bbooton@fau.edu (B. Booton), sagher@fau.edu (Y. Sagher).

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then

$$\|\{\lambda_n\}\|_{l(q',q)} \leqslant C\|f\|_{L^q}.$$

Conversely, if $\|\{\lambda_n\}\|_{l(q',q)} < \infty$ and $1 < q < \infty$, then f, defined as $f(x) = \sum \lambda_n^* \cos nx$, satisfies $\|f\|_q \leqslant C\|\{\lambda_n\}\|_{l(q',q)}$.

These inequalities were precursors to the Lorentz spaces $L(p,q)$ ($l(p,q)$ for sequences), defined in full generality by G.G. Lorentz [10], A.P. Calderón [3], and R. Hunt [8]. Note that $(\sum_{n=1}^{\infty}(\lambda_n^*)^q n^{q-2})^{1/q}$ can be taken as the $l(q',q)$ norm of $\{\lambda_n\}$, and so (1) can be written as $\|f\|_{L^q(0,\pi)} \leqslant C\|\{\lambda_n\}\|_{l(q',q)}$ for $q \geqslant 2$.

Paley [11] generalized some of Hardy and Littlewood's results on trigonometric series to ones on more general bounded orthonormal systems on $(0,\pi)$. Specifically, for $q \geqslant 2$, if $\{\phi_n\}$ is a bounded orthonormal system on $(0,\pi)$, and $\sum(\lambda_n^*)^q n^{q-2} < \infty$, then $f = \sum \lambda_n \phi_n \in L^q(0,\pi)$, and $\|f\|_{L^q(0,\pi)} \leqslant C\|\{\lambda_n\}\|_{l(q',q)}$. Also, for $1 < q \leqslant 2$, if $\{\phi_n\}$ is a bounded orthonormal system on $(0,\pi)$, and $f = \sum \lambda_n \phi_n \in L^q(0,\pi)$, then $\sum(\lambda_n^*)^q n^{q-2} < \infty$, and $\|\{\lambda_n\}\|_{l(q',q)} \leqslant C\|f\|_{L^q(0,\pi)}$. Applying Paley's theorem to appropriate permutations of the trigonometric system gives the Hardy and Littlewood theorem.

Zygmund [14] introduced interpolation methods into this subject by proving Paley's theorem using the Marcinkiewicz interpolation theorem. Work along these lines was continued in [12,13]. Note that classes of functions with nonincreasing Fourier transforms do not form linear spaces or even groups, and so the application of interpolation theory required recasting the theory to interpolated semi-groups.

Many additional results may be found in Boas [2]. Boas's presentation in [2] does not use interpolation techniques. We will show how results in [2] can be strengthened, and their proofs simplified, using interpolation theory. We will also extend results that Boas proved for monotone functions to quasi-monotone ones.

2. Inequalities for series and integrals

We recall the definition of $L(p,q)$ spaces.

Definition 1. Let (M, Σ, μ) be a σ-finite measure space. Assume f is a measurable function and that $\lim_{a\to\infty} \mu\{|f| > a\} = 0$. Let f^* denote the nonincreasing rearrangement of $|f|$. For $0 < p < \infty$, $0 < q \leqslant \infty$, define

$$\|f\|_{L(p,q)} = \left(\int_0^{\mu(M)} \left(x^{\frac{1}{p}} f^*(x)\right)^q \frac{dx}{x} \right)^{\frac{1}{q}}. \tag{2}$$

In the special case where (M, Σ, μ) is the space of positive integers with $\mu(\{n\}) = 1$, we use an equivalent norm:

$$\|\{a_n\}\|_{l(p,q)} = \left(\sum_{n=1}^{\infty} \frac{1}{n} \left(n^{\frac{1}{p}} a_n^*\right)^q \right)^{\frac{1}{q}}.$$

We define $L(p,q) = \{f\colon \|f\|_{L(p,q)} < \infty\}$ and $l(p,q) = \{\{a_n\}\colon \|\{a_n\}\|_{l(p,q)} < \infty\}$.

Definition 2. Let (M, Σ, μ) be a measure space. Let ω be a weight function; i.e., an a.e. positive measurable function. For $0 < p \leqslant \infty$ and f measurable, define $\|f\|_{L^p_\omega} = \|\omega f\|_{L^p}$ and $L^p_\omega = \{f\colon \|f\|_{L^p_\omega} < \infty\}$.

It will be convenient to write $\omega(p,q)$ for the weight $\omega(x) = x^{\frac{1}{p}-\frac{1}{q}}$ on $\mathbf{R}^+$.

In the sequel we shall apply interpolation with change of measure.

Theorem 1. *(See [4].) Assume that* $0 < \theta < 1$, $0 < p, q \leqslant \infty$. *Then*

$$\|f\|_{(L^p, L^p_\omega)_{\theta,q}} \sim \max\left\{ \left(\int_0^\infty \left(t^{-\theta}\|f \cdot t\omega I_{\{t\omega<1\}}\|_{L^p}\right)^q \frac{dt}{t} \right)^{\frac{1}{q}}, \left(\int_0^\infty \left(t^{-\theta}\|f I_{\{t\omega\geqslant 1\}}\|_{L^p}\right)^q \frac{dt}{t} \right)^{\frac{1}{q}} \right\}. \tag{3}$$

The statement of Theorem 1 in [4] is for $1 \leqslant p, q \leqslant \infty$. The extension for $0 < p, q \leqslant \infty$ is routine and so we shall skip its proof.

Lemma 1. *(See [7].) Assume that* (M, Σ, μ) *is a measure space, and that* $0 \leqslant f, g$ *are measurable. For any* $E \in \Sigma$,

$$\int_E fg\, d\mu \leqslant \int_0^{\mu(E)} f^* g^*.$$

As a consequence, we have that $\|f\|_{L^q_{\omega(p,q)}} \leqslant \|f\|_{L(p,q)}$ for $q \leqslant p$ and $\|f\|_{L(p,q)} \leqslant \|f\|_{L^q_{\omega(p,q)}}$ for $q \geqslant p$.

We will make frequent use of Hardy's inequalities. These inequalities have several equivalent formulations and for the convenience of the reader, we include the forms that we shall use.

Throughout, when the domain of a function is not specified, we take it to be $(0, a)$, where $0 < a \leqslant \infty$.

Theorem 2. *Assume that* $f \geqslant 0$, $\alpha > 0$, *and* $1 \leqslant q \leqslant \infty$. *Then*

$$\left(\int_0^a \left(x^{-\alpha} \int_0^x f(t)\frac{dt}{t} \right)^q \frac{dx}{x} \right)^{\frac{1}{q}} \leqslant \frac{1}{\alpha} \left(\int_0^a \left(x^{-\alpha} f(x)\right)^q \right)^{\frac{1}{q}} \tag{4}$$

and

$$\left(\int_0^a \left(x^{\alpha} \int_x^a f(t)\frac{dt}{t} \right)^q \frac{dx}{x} \right)^{\frac{1}{q}} \leqslant \frac{1}{\alpha} \left(\int_0^a \left(x^{\alpha} f(x)\right)^q \right)^{\frac{1}{q}}. \tag{5}$$

Hardy, Littlewood, and Paley considered norm inequalities for the Fourier transforms when the functions or the Fourier transforms are monotone. It turns out that their results also hold for quasi-monotone functions. This generalization is particularly interesting when one considers Besov norm inequalities, since the class of functions with quasi-monotone Fourier transforms is closed under differentiation; see [12].

Definition 3. Define for $\beta \geqslant 0$: $f \in QD(\beta)$ if $x^{-\beta} f(x) \searrow 0$; and $f \in QD$ if $f \in QD(\beta)$ for some $\beta \geqslant 0$. Similarly, $\{a_n\} \in QD(\beta)$ if $n^{-\beta} a_n \searrow 0$, and $\{a_n\} \in QD$ if $\{a_n\} \in QD(\beta)$ for some $\beta \geqslant 0$.

Theorem 3. *Assume that $f \in QD(\beta)$. For $0 < q \leqslant \infty$, denote $B_q(t) = (\int_0^t (f(s))^q \frac{ds}{s})^{1/q}$ and $D_q(t) = (\int_t^a (f(s))^q \frac{ds}{s})^{1/q}$. Then for $0 < q < r \leqslant \infty$,*

$$f(t) \leqslant C(\beta, r) B_r(t) \leqslant C(\beta, r, q) B_q(t) \tag{6}$$

and

$$f(t) \leqslant C(\beta, r) D_r\left(\frac{t}{2}\right) \leqslant C(\beta, r, q) D_q\left(\frac{t}{4}\right). \tag{7}$$

Proof.

$$B_q(t) = \left(\int_0^t \left(s^\beta s^{-\beta} f(s)\right)^q \frac{ds}{s}\right)^{\frac{1}{q}} \geqslant t^{-\beta} f(t) \left(\int_0^t s^{\beta q} \frac{ds}{s}\right)^{\frac{1}{q}} = (\beta q)^{-\frac{1}{q}} f(t).$$

Hence,

$$\begin{aligned} B_r(t) &= \left(\int_0^t \left(f(s)\right)^{r-q} \left(f(s)\right)^q \frac{ds}{s}\right)^{\frac{1}{r}} \\ &\leqslant (\beta q)^{\frac{1}{q}-\frac{1}{r}} \left(\int_0^t \left(B_q(s)\right)^{r-q} \left(f(s)\right)^q \frac{ds}{s}\right)^{\frac{1}{r}} \\ &\leqslant (\beta q)^{\frac{1}{q}-\frac{1}{r}} \left(B_q(t)\right)^{1-\frac{q}{r}} \left(\int_0^t \left(f(s)\right)^q \frac{ds}{s}\right)^{\frac{1}{r}} = (\beta q)^{\frac{1}{q}-\frac{1}{r}} B_q(t) \end{aligned}$$

proving (6). Similarly,

$$\begin{aligned} D_q\left(\frac{t}{2}\right) &= \left(\int_{\frac{t}{2}}^a \left(f(s)\right)^q \frac{ds}{s}\right)^{\frac{1}{q}} \geqslant \left(\int_{\frac{t}{2}}^t \left(s^\beta s^{-\beta} f(s)\right)^q \frac{ds}{s}\right)^{\frac{1}{q}} \\ &\geqslant t^{-\beta} f(t) \left(\int_{\frac{t}{2}}^t s^{\beta q} \frac{ds}{s}\right)^{\frac{1}{q}} = C(\beta, q) f(t). \end{aligned}$$

Therefore,

$$\begin{aligned} D_r(t) &= \left(\int_t^a \left(f(s)\right)^{r-q} \left(f(s)\right)^q \frac{ds}{s}\right)^{\frac{1}{r}} \\ &\leqslant C(\beta, q, r) \left(\int_t^a \left(D_q\left(\frac{s}{2}\right)\right)^{r-q} \left(f(s)\right)^q \frac{ds}{s}\right)^{\frac{1}{r}} \\ &\leqslant C(\beta, q, r) \left(D_q\left(\frac{t}{2}\right)\right)^{1-\frac{q}{r}} \left(\int_t^a \left(f(s)\right)^q \frac{ds}{s}\right)^{\frac{1}{r}} \\ &= C(\beta, q, r) \left(D_q\left(\frac{t}{2}\right)\right)^{1-\frac{q}{r}} \left(D_q(t)\right)^{\frac{q}{r}} \leqslant C(\beta, q, r) D_q\left(\frac{t}{2}\right). \end{aligned}$$

□

Assume that f and g are two nonnegative functions. We write $f \sim g$ if there exists a constant C so that for all x,

$$\frac{1}{C} f(x) \leqslant g(x) \leqslant C f(x).$$

Theorem 4. *Assume that $f \in QD$, $0 < q \leqslant \infty$, $\alpha > 0$. Then*

$$\left(\int_0^a \left(t^{-\alpha} \int_0^t f(s) \frac{ds}{s} \right)^q \frac{dt}{t} \right)^{\frac{1}{q}} \sim \left(\int_0^a \left(t^{-\alpha} f(t) \right)^q \frac{dt}{t} \right)^{\frac{1}{q}} \tag{8}$$

and

$$\left(\int_0^a \left(t^{\alpha} \int_t^a f(s) \frac{ds}{s} \right)^q \frac{dt}{t} \right)^{\frac{1}{q}} \sim \left(\int_0^a \left(t^{\alpha} f(t) \right)^q \frac{dt}{t} \right)^{\frac{1}{q}}. \tag{9}$$

Proof. Let us show that there exists a constant $C = C(\alpha, \beta, q)$ so that for all $f \in QD(\beta)$,

$$\left(\int_0^a \left(t^{-\alpha} \int_0^t f(s) \frac{ds}{s} \right)^q \frac{dt}{t} \right)^{\frac{1}{q}} \leqslant C \left(\int_0^a \left(t^{-\alpha} f(t) \right)^q \frac{dt}{t} \right)^{\frac{1}{q}}.$$

For $1 \leqslant q \leqslant \infty$ this is (4). For $0 < q < 1$, using (7),

$$\int_0^a \left(t^{-\alpha} \int_0^t f(s) \frac{ds}{s} \right)^q \frac{dt}{t} = \int_0^a \left(t^{-\alpha} B_1(t) \right)^q \frac{dt}{t} \leqslant C \int_0^a \left(t^{-\alpha} B_q(t) \right)^q \frac{dt}{t}$$

$$= C \int_0^a \left(f(s) \right)^q \left(\int_s^a t^{-\alpha q} \frac{dt}{t} \right) \frac{ds}{s} \leqslant C \int_0^a \left(s^{-\alpha} f(s) \right)^q \frac{ds}{s}.$$

Conversely, by (6),

$$\left(\int_0^a \left(t^{-\alpha} f(t) \right)^q \frac{dt}{t} \right)^{\frac{1}{q}} \leqslant \beta \left(\int_0^a \left(t^{-\alpha} B_1(t) \right)^q \frac{dt}{t} \right)^{\frac{1}{q}} = \beta \left(\int_0^a \left(t^{-\alpha} \int_0^t f(s) \frac{ds}{s} \right)^q \frac{dt}{t} \right)^{\frac{1}{q}}$$

proving (8). Note that the preceding inequality holds for all $\alpha \in \mathbf{R}$.

Let us see that for all $f \in QD(\beta)$,

$$\left(\int_0^a \left(t^{\alpha} \int_t^a f(s) \frac{ds}{s} \right)^q \frac{dt}{t} \right)^{\frac{1}{q}} \leqslant C \left(\int_0^a \left(t^{\alpha} f(t) \right)^q \frac{dt}{t} \right)^{\frac{1}{q}}.$$

For $1 \leqslant q \leqslant \infty$ this is (5). For $0 < q < 1$, using (7),

$$\left(\int_0^a \left(t^\alpha \int_t^a f(s)\frac{ds}{s}\right)^q \frac{dt}{t}\right)^{\frac{1}{q}}$$
$$= \left(\int_0^a \left(t^\alpha D_1(t)\right)^q \frac{dt}{t}\right)^{\frac{1}{q}} \leqslant C\left(\int_0^a \left(t^\alpha D_q\left(\frac{t}{2}\right)\right)^q \frac{dt}{t}\right)^{\frac{1}{q}}$$
$$= C\left(\int_0^a t^{\alpha q}\left(\int_{\frac{t}{2}}^a \left(f(s)\right)^q \frac{ds}{s}\right)\frac{dt}{t}\right)^{\frac{1}{q}} \leqslant C\left(\int_0^a \left(f(s)\right)^q \left(\int_0^{2s} t^{\alpha q}\frac{dt}{t}\right)\frac{ds}{s}\right)^{\frac{1}{q}}$$
$$= C\left(\int_0^a \left(s^\alpha f(s)\right)^q \frac{ds}{s}\right)^{\frac{1}{q}}.$$

Conversely, by (6),

$$\left(\int_0^a \left(t^\alpha f(t)\right)^q \frac{dt}{t}\right)^{\frac{1}{q}} \leqslant C\left(\int_0^a \left(t^\alpha D_1\left(\frac{t}{2}\right)\right)^q \frac{dt}{t}\right)^{\frac{1}{q}} = C\left(\int_0^{\frac{a}{2}} \left(t^\alpha \int_t^a f(s)\frac{ds}{s}\right)^q \frac{dt}{t}\right)^{\frac{1}{q}}$$
$$\leqslant C\left(\int_0^a \left(t^\alpha \int_t^a f(s)\frac{ds}{s}\right)^q \frac{dt}{t}\right)^{\frac{1}{q}}$$

proving (9). Observe that this inequality holds for all $\alpha \in \mathbf{R}$. □

Our next goal is to show that a *QD* function can be written as the difference of two non-increasing functions whose $L^q_{\omega(p,q)}$ and $L(p,q)$ norms are equivalent to those of the original function.

From the corresponding inequality for the sum we get:

Lemma 2. *Assume that f, g are measurable. For $a, b \geqslant 0$,*

$$(fg)^*(a+b) \leqslant f^*(a)g^*(b).$$

Corollary 1. *Assume that $f \in QD(\beta)$ and is right continuous. For $s \geqslant t$,*

$$s^{-\beta} f(s) = \left(u^{-\beta} f(u)\right)^*(s) \leqslant t^{-\beta} f^*(s-t).$$

In particular,

$$f(s) \leqslant 2^\beta f^*\left(\frac{s}{2}\right). \tag{10}$$

The next two theorems were proved in [13] for $1 < p < \infty$ and $1 \leqslant q \leqslant \infty$.

Theorem 5. *Assume that $f \in QD$, and that $\int_x^a f(t)\frac{dt}{t} < \infty$ for all $0 < x \leqslant a$. Then there exist nonincreasing functions f_1, f_2 so that $f = f_2 - f_1$.*

If furthermore $f \in L^q_{\omega(p,q)}$, $0 < p < \infty$, $0 < q \leqslant \infty$, *then for* $j = 1, 2$,

$$\|f_j\|_{L^q_{\omega(p,q)}} \sim \|f\|_{L^q_{\omega(p,q)}}. \tag{11}$$

Proof. We define $f_1(x) = \beta \int_x^a f(t)\frac{dt}{t}$ and $f_2 = f + f_1$. Let us see that f_2 is nonincreasing. Let $0 < x_0 < x_1$. Then

$$\begin{aligned} f_2(x_0) - f_2(x_1) &= f(x_0) - f(x_1) + \beta \int_{x_0}^{x_1} f(t)\frac{dt}{t} \\ &= f(x_0) - f(x_1) + \beta \int_{x_0}^{x_1} \big(t^{-\beta} f(t)\big) t^{\beta} \frac{dt}{t} \\ &\geqslant f(x_0) - f(x_1) + \beta x_1^{-\beta} f(x_1) \int_{x_0}^{x_1} t^{\beta} \frac{dt}{t} \\ &= f(x_0) - \left(\frac{x_0}{x_1}\right)^{\beta} f(x_1) \geqslant 0. \end{aligned}$$

Assume that $f \in L^q_{\omega(p,q)}$. From (9) it follows that $\int_x^a f(t)\frac{dt}{t} < \infty$ for all $0 < x \leqslant a$, and furthermore $\|f_1\|_{L^q_{\omega(p,q)}} \sim \|f\|_{L^q_{\omega(p,q)}}$, and thus

$$\|f\|_{L^q_{\omega(p,q)}} \leqslant \|f_2\|_{L^q_{\omega(p,q)}} \leqslant C\big(\|f\|_{L^q_{\omega(p,q)}} + \|f_1\|_{L^q_{\omega(p,q)}}\big) \leqslant C\|f\|_{L^q_{\omega(p,q)}}$$

so that $\|f\|_{L^q_{\omega(p,q)}} \sim \|f_2\|_{L^q_{\omega(p,q)}}$. □

Theorem 6. *Assume that* $f \in QD$, $0 < p < \infty$, *and* $0 < q \leqslant \infty$. *Then*

$$\|f\|_{L(p,q)} \sim \|f\|_{L^q_{\omega(p,q)}}.$$

Proof. For $a < \infty$, we extend f by defining $f(x) = 0$ for $x \geqslant a$. Assume that $f \in QD(\beta)$, and let $g(x) = \lim_{t \to x^+}(t^{-\beta} f(t))$. Then g is right-continuous and nonincreasing. Therefore, $h(x) := x^{\beta} g(x) \in QD$ and is right-continuous, and $x^{\beta} g(x) = f(x)$ nearly everywhere, so that

$$\|f\|_{L(p,q)} = \|h\|_{L(p,q)} \quad \text{and} \quad \|f\|_{L^q_{\omega(p,q)}} = \|h\|_{L^q_{\omega(p,q)}}.$$

We can assume therefore that f is right-continuous. By (10),

$$\|f\|_{L^q_{\omega(p,q)}} = \left(\int_0^{\infty} \big(x^{\frac{1}{p}} f(x)\big)^q \frac{dx}{x}\right)^{\frac{1}{q}} \leqslant 2^{\beta} \left(\int_0^{\infty} \left(x^{\frac{1}{p}} f^*\left(\frac{x}{2}\right)\right)^q \frac{dx}{x}\right)^{\frac{1}{q}} = 2^{\beta + \frac{1}{p}} \|f\|_{L(p,q)}.$$

For the other inequality, let f_1, f_2 be as in Theorem 5. Then

$$\|f\|_{L(p,q)} \leqslant \|f_2\|_{L(p,q)} = \|f_2\|_{L^q_{\omega(p,q)}} \leqslant C\|f\|_{L^q_{\omega(p,q)}}. \qquad \square$$

Corollary 2. *Assume that $f \in L(p,q) \cap QD$, where $0 < p < \infty$ and $0 < q \leqslant \infty$. Then there exist nonincreasing functions f_1, f_2 so that $f = f_2 - f_1$ and for $j = 1, 2$,*

$$\|f_j\|_{L(p,q)} \sim \|f\|_{L(p,q)}. \tag{12}$$

Clearly Theorem 5 and Corollary 2 hold for sequence spaces as well.

The series version of the following lemma, with $\eta = \delta$, is proved differently in [2, Lemma 6.18].

For $0 < q \leqslant \infty$, we define $q' = \frac{q}{q-1}$, so that $\frac{1}{q} + \frac{1}{q'} = 1$. Note that for $0 < q < 1$, $q' < 0$.

Lemma 3. *Assume that G is nondecreasing and right-continuous on (a,b), where $0 \leqslant a < b \leqslant \infty$. For $s > 0$, $\delta > 0$, $0 < \eta \leqslant \delta$, and $1 \leqslant q \leqslant \infty$,*

$$\left(\int_a^b \left(x^{-\delta} \int_{(a,x]} y^s \, dG(y)\right)^q \frac{dx}{x}\right)^{\frac{1}{q}}$$
$$\leqslant C(s,\eta,q)\left(\int_a^b \left(x^{s-\eta} \int_{(x,b)} y^{\eta-\delta} \, dG(y)\right)^q \frac{dx}{x}\right)^{\frac{1}{q}} + \frac{K(a)a^s}{(\delta q)^{\frac{1}{q}}} \int_{(a,b)} y^{-\delta} \, dG(y) \tag{13}$$

and

$$\left(\int_a^b \left(x^{\delta} \int_{[x,b)} dG(y)\right)^q \frac{dx}{x}\right)^{\frac{1}{q}}$$
$$\leqslant C(s,\eta,q)\left(\int_a^b \left(x^{\eta-s} \int_{(a,x)} y^{s+\delta-\eta} \, dG(y)\right)^q \frac{dx}{x}\right)^{\frac{1}{q}} + \frac{K(\frac{1}{b})}{(\delta q)^{\frac{1}{q}} b^s} \int_{(a,b)} y^{s+\delta} \, dG(y), \tag{14}$$

where $K(0) = 0$ and $K(a) = 1$ for $a > 0$. If $a = 0$, then (13) *holds for $0 < q \leqslant \infty$, and if $b = \infty$, then* (14) *holds for $0 < q \leqslant \infty$. In both* (13) *and* (14)*, if $q \geqslant 1$, then $C(s,\eta,q) \leqslant \frac{s}{\eta}$.*

Proof. It suffices to consider $q < \infty$.

$$\left(\int_a^b \left(x^{-\delta} \int_{(a,x]} y^s \, dG(y)\right)^q \frac{dx}{x}\right)^{\frac{1}{q}}$$
$$= \left(\int_a^b \left(x^{-\delta} \int_{(a,x]} \left(s \int_a^y t^s \frac{dt}{t} + K(a)a^s\right) dG(y)\right)^q \frac{dx}{x}\right)^{\frac{1}{q}}$$
$$\leqslant s\left(\int_a^b \left(x^{-\delta} \int_{(a,x]} \left(\int_a^y t^s \frac{dt}{t}\right) dG(y)\right)^q \frac{dx}{x}\right)^{\frac{1}{q}}$$
$$+ K(a)a^s \left(\int_a^b \left(x^{-\delta} \int_{(a,x]} dG(y)\right)^q \frac{dx}{x}\right)^{\frac{1}{q}}.$$

We consider each of the last two terms.

$$
\begin{aligned}
&s\left(\int_a^b\left(x^{-\delta}\int_{(a,x]}\left(\int_a^y t^s\frac{dt}{t}\right)dG(y)\right)^q\frac{dx}{x}\right)^{\frac{1}{q}}\\
&\quad= s\left(\int_a^b\left(x^{-\delta}\int_a^x t^s\left(\int_{(t,x]}dG(y)\right)\frac{dt}{t}\right)^q\frac{dx}{x}\right)^{\frac{1}{q}}\\
&\quad= s\left(\int_a^b\left(x^{-\delta}\int_a^x t^s\left(\int_{(t,x]}y^{\delta-\eta}y^{\eta-\delta}\,dG(y)\right)\frac{dt}{t}\right)^q\frac{dx}{x}\right)^{\frac{1}{q}}\\
&\quad\leqslant s\left(\int_a^b\left(x^{-\eta}\int_a^x t^s\left(\int_{(t,x]}y^{\eta-\delta}\,dG(y)\right)\frac{dt}{t}\right)^q\frac{dx}{x}\right)^{\frac{1}{q}}\\
&\quad\leqslant s\left(\int_0^b\left(x^{-\eta}\int_0^x t^s I_{(a,b)}(t)\left(\int_{(t,b)}y^{\eta-\delta}\,dG(y)\right)\frac{dt}{t}\right)^q\frac{dx}{x}\right)^{\frac{1}{q}}.
\end{aligned}
$$

We apply Hardy's inequality to the last term. The case $q\geqslant 1$ requires no comment. If $a=0$,

$$
t^s\int_{(t,b)}y^{\eta-\delta}\,dG(y)\in QD(s)
$$

and so by (8), the application of Hardy's inequality is valid for $0<q\leqslant\infty$. Thus,

$$
\begin{aligned}
&s\left(\int_0^b\left(x^{-\eta}\int_0^x t^s I_{(a,b)}(t)\left(\int_{(t,b)}y^{\eta-\delta}\,dG(y)\right)\frac{dt}{t}\right)^q\frac{dx}{x}\right)^{\frac{1}{q}}\\
&\quad\leqslant C(s,\eta,q)\left(\int_0^b\left(x^{s-\eta}I_{(a,b)}(x)\left(\int_{(x,b)}y^{\eta-\delta}\,dG(y)\right)\right)^q\frac{dx}{x}\right)^{\frac{1}{q}}\\
&\quad= C(s,\eta,q)\left(\int_a^b\left(x^{s-\eta}\int_{(x,b)}y^{\eta-\delta}\,dG(y)\right)^q\frac{dx}{x}\right)^{\frac{1}{q}}.
\end{aligned}
$$

Also, by Minkowski's inequality,

$$
\begin{aligned}
K(a)a^s\left(\int_a^b\left(x^{-\delta}\int_{(a,x]}dG(y)\right)^q\frac{dx}{x}\right)^{\frac{1}{q}}&\leqslant K(a)a^s\int_{(a,b)}\left(\int_y^b x^{-\delta q}\frac{dx}{x}\right)^{\frac{1}{q}}dG(y)\\
&\leqslant(\delta q)^{-\frac{1}{q}}K(a)a^s\int_{(a,b)}y^{-\delta}\,dG(y)
\end{aligned}
$$

proving (13).

We use $-v^{-s}\,dG(\frac{1}{v})$ as the measure in (13) to prove (14),

$$\begin{aligned}
&\left(\int_a^b \left(x^\delta \int_{[x,b)} dG(y)\right)^q \frac{dx}{x}\right)^{\frac{1}{q}} \\
&= \left(\int_{\frac{1}{b}}^{\frac{1}{a}} \left(u^{-\delta} \int_{(\frac{1}{b},u]} \left(-v^s v^{-s}\, dG\left(\frac{1}{v}\right)\right)\right)^q \frac{du}{u}\right)^{\frac{1}{q}} \\
&\leqslant C(s,\eta,q)\left(\int_{\frac{1}{b}}^{\frac{1}{a}} \left(u^{s-\eta} \int_{(u,\frac{1}{a})} v^{\eta-\delta}\left(-v^{-s}\, dG\left(\frac{1}{v}\right)\right)\right)^q \frac{du}{u}\right)^{\frac{1}{q}} \\
&\quad + \frac{K(\frac{1}{b})}{(\delta q)^{\frac{1}{q}} b^s} \int_{(\frac{1}{b},\frac{1}{a})} \left(-v^{-\delta} v^{-s}\, dG\left(\frac{1}{v}\right)\right) \\
&= C(s,\eta,q)\left(\int_a^b \left(x^{\eta-s} \int_{(a,x)} y^{\delta+s-\eta}\, dG(y)\right)^q \frac{dx}{x}\right)^{\frac{1}{q}} \\
&\quad + \frac{K(\frac{1}{b})}{(\delta q)^{\frac{1}{q}} b^s} \int_{(a,b)} y^{\delta+s}\, dG(y). \qquad \square
\end{aligned}$$

3. Norm inequalities for Fourier transforms

In this section, $\mathbf{I}$ denotes the interval $(0,\pi)$, $E(x) = \sum_{n=1}^\infty a_n \cos nx$, and $S(x) = \sum_{n=1}^\infty a_n \sin nx$.

The following is a generalization to quasi-monotone sequences of a theorem of Hardy and Littlewood for monotone sequences.

Theorem 7. *(See [12,13].) Assume that f is either the Fourier sine or Fourier cosine series associated with $\{a_n\}$. If either $\{a_n\} \in QD$ or $f \in QD$ then for $1 < p < \infty$ and $0 < q \leqslant \infty$,*

$$\left\|\{a_n\}\right\|_{l(p',q)} \sim \|f\|_{L(p,q)(\mathbf{I})} \tag{15}$$

and for $1 < p < \infty$ and $1 \leqslant q \leqslant \infty$,

$$\left\|\{a_n\}\right\|_{l^q_{\omega(p',q)}} \sim \|f\|_{L^q_{\omega(p,q)}(\mathbf{I})}. \tag{16}$$

Theorem 8. *(See [9].) Let H denote the Hilbert transform. Assume that for some $\alpha \in \mathbf{R}$, we have $\frac{f(x)-\alpha}{x} \in L^1(\mathbf{I})$. Then*

$$H\left(\frac{f(x)-\alpha}{x}\right) = \frac{Hf(x) - Hf(0)}{x}.$$

Definition 4. (See [2].) For f so that $xf(x) \in L^1(\mathbf{I})$, define the sequence of generalized sine coefficients $\{b_n\}$ of f by

$$b_n = \frac{2}{\pi} \int_0^\pi f(x) \sin nx \, dx.$$

For f so that $x^2 f(x) \in L^1(\mathbf{I})$, define the sequence of generalized cosine coefficients $\{a_n\}$ of f by

$$a_n = \frac{2}{\pi} \int_0^\pi f(x)(1 - \cos nx) \, dx.$$

The case $s = 1$ in the next theorem is Theorem 6.11 in [2].

Theorem 9. *Assume that $g \geqslant 0$ on $\mathbf{I}$, $xg(x) \in L^1(\mathbf{I})$, and $1 < p < \infty$. Assume that $\{\lambda_n\}$ is the sequence of generalized sine coefficients or cosine coefficients of g. For $\frac{1}{p} < s$ and $0 < q \leqslant \infty$,*

$$\left\| \frac{1}{x^s} \int_0^x t^s g(t) \, dt \right\|_{L^q_{\omega(p,q)}(\mathbf{I})} \leqslant C \left\| \left\{ \frac{\lambda_n}{n} \right\} \right\|_{l(p',q)} \tag{17}$$

and for $1 \leqslant q \leqslant \infty$,

$$\left\| \frac{1}{x^s} \int_0^x t^s g(t) \, dt \right\|_{L^q_{\omega(p,q)}(\mathbf{I})} \leqslant C \left\| \left\{ \frac{\lambda_n}{n} \right\} \right\|_{l^q_{\omega(p',q)}}. \tag{18}$$

For $s > 0$ and $1 \leqslant q \leqslant \infty$,

$$\left\| \left\{ \frac{\lambda_n}{n} \right\} \right\|_{l(p',q)} \leqslant C \left(\left\| \frac{1}{x^s} \int_0^x t^s g(t) \, dt \right\|_{L^q_{\omega(p,q)}(\mathbf{I})} + \int_0^\pi t^{s+\frac{1}{p}} g(t) \, dt \right) \tag{19}$$

and

$$\left\| \left\{ \frac{\lambda_n}{n} \right\} \right\|_{l^q_{\omega(p',q)}} \leqslant C \left(\left\| \frac{1}{x^s} \int_0^x t^s g(t) \, dt \right\|_{L^q_{\omega(p,q)}(\mathbf{I})} + \int_0^\pi t^{s+\frac{1}{p}} g(t) \, dt \right). \tag{20}$$

Proof. Let $G(x) = \int_x^\pi g(t) \, dt$. Note that $G \in L^1(\mathbf{I})$. If λ_n are the generalized sine coefficients of g, then $\frac{\lambda_n}{n}$ are the cosine coefficients of G, and if λ_n are the generalized cosine coefficients of g then $\frac{\lambda_n}{n}$ are the sine coefficients of G.

By (13), for $0 < q \leqslant \infty$,

$$\left\| \frac{1}{x^s} \int_0^x t^s g(t) \, dt \right\|_{L^q_{\omega(p,q)}(\mathbf{I})} \leqslant C \|G\|_{L^q_{\omega(p,q)}(\mathbf{I})} = C \|G\|_{L(p,q)(\mathbf{I})}.$$

By Theorem 7, since $G \searrow$, for $1 < p < \infty$, $0 < q \leqslant \infty$,

$$\|G\|_{L(p,q)(\mathbf{I})} \sim \left\| \left\{ \frac{\lambda_n}{n} \right\} \right\|_{l(p',q)}$$

and for $1 < p \leqslant \infty$, $1 \leqslant q \leqslant \infty$,

$$\|G\|_{L^q_{\omega(p,q)}(\mathbf{I})} \sim \left\| \left\{ \frac{\lambda_n}{n} \right\} \right\|_{l^q_{\omega(p',q)}}$$

proving (17) and (18).

By (14), for $1 \leqslant q \leqslant \infty$,

$$\|G\|_{L^q_{\omega(p,q)}(\mathbf{I})} \leqslant ps \left\| \frac{1}{x^s} \int_0^x t^s g(t)\,dt \right\|_{L^q_{\omega(p,q)}(\mathbf{I})} + \left(\frac{p}{q}\right)^{\frac{1}{q}} \frac{1}{\pi^s} \int_0^\pi t^{s+\frac{1}{p}} g(t)\,dt$$

proving (19) and (20). □

The following two theorems are related to Theorem 7 in [1]. We state that theorem using the notation of this paper.

If $\lambda_n \geqslant 0$ are the Fourier sine or cosine coefficients of f, then for $0 \leqslant a < \pi$, $1 < p < \infty$ and $1 < q < \infty$,

$$\frac{1}{C} \left\| \frac{f(x) - f(a)}{x - a} \right\|_{L^q_{|x-a|^{\frac{1}{p}-\frac{1}{q}}}} \leqslant \left\| \left\{ \sum_{k=n}^{\infty} \lambda_k \right\} \right\|_{l^q_{\omega(p',q)}} \leqslant C \left\| \frac{f(x)}{x} \right\|_{L^q_{\omega(p,q)}}.$$

We generalize this theorem in several ways. We relax the condition $\lambda_n \geqslant 0$, extend the range of the parameter q, and prove related $L(p,q)$ inequalities.

Given a sequence $\{\lambda_k\}$ so that $\sum_{k=1}^{\infty} \lambda_k$ converges, we denote $\Lambda_n = \sum_{k=n}^{\infty} \lambda_k$. A summation by parts argument proves the following:

Lemma 4. *Assume that* $\sum_{n=1}^{\infty} \lambda_n$, $\sum_{n=1}^{\infty} \lambda_n \sin nx$, *and* $\sum_{n=1}^{\infty} \lambda_n \cos nx$ *converge. Then for* $0 < x \leqslant \pi$,

$$\sum_{n=1}^{\infty} \Lambda_n \cos nx = \frac{\cos\frac{x}{2}}{2\sin\frac{x}{2}} \sum_{n=1}^{\infty} \lambda_n \sin nx - \frac{1}{2} \sum_{n=1}^{\infty} \lambda_n (1 - \cos nx)$$

and

$$\sum_{n=1}^{\infty} \Lambda_n \sin nx = \frac{\cos\frac{x}{2}}{2\sin\frac{x}{2}} \sum_{n=1}^{\infty} \lambda_n (1 - \cos nx) + \frac{1}{2} \sum_{n=1}^{\infty} \lambda_n \sin nx.$$

Lemma 5. *Assume that* $\sum_{n=1}^{\infty} \lambda_n$ *converges,* $\{\Lambda_n\} \in QD$, *and* $\sum_{n=1}^{\infty} \frac{\Lambda_n}{n} < \infty$. *Then* $\sum_{n=1}^{\infty} \lambda_n \sin nx$ *and* $\sum_{n=1}^{\infty} \lambda_n \cos nx$ *converge.*

Proof. By the decomposition of QD sequences as in Theorem 5, there exist $\{\Lambda_{1,n}\}$, $\{\Lambda_{2,n}\}$ with $\Lambda_{j,n} \searrow 0$ so that $\Lambda_n = \Lambda_{1,n} - \Lambda_{2,n}$. Since $\Lambda_{j,n} \searrow 0$, $\sum_{n=1}^{\infty} \Lambda_{j,n} \sin nx$ converge. Therefore, $\sum_{n=1}^{\infty} \Lambda_n \sin nx$ converges, as does $\sum_{n=1}^{\infty} \Lambda_{n+1} \sin nx$. Therefore, $\sum_{n=1}^{\infty} \lambda_n \sin nx = \sum_{n=1}^{\infty} (\Lambda_n - \Lambda_{n+1}) \sin nx$ converges. The proof for the cosine series is the same. □

Theorem 10.

(a) *Assume that* $\frac{f(x)}{x} \in L(p,q)(\mathbf{I})$, $1 < p < \infty$. *Let* $\{\lambda_k\}$ *be the sequence of sine coefficients of* f. *Then* $\sum_{k=1}^{\infty} \lambda_k$ *converges. Assume that* $\{\Lambda_n\} \in QD$. *Then for* $0 < q \leqslant \infty$,

$$\big\|\{\Lambda_n\}\big\|_{l(p',q)} \leqslant C \left\| \frac{f(x)}{x} \right\|_{L(p,q)(\mathbf{I})} \tag{21}$$

and for $1 < q < \infty$,

$$\big\|\{\Lambda_n\}\big\|_{l(p',q)} \leqslant C \left\| \frac{f(x)}{x} \right\|_{L^q_{\omega(p,q)}(\mathbf{I})}. \tag{22}$$

(b) *Assume that for some* $A \in \mathbf{R}$, $\frac{f(x)-A}{x} \in L(p,q)(\mathbf{I})$, $1 < p < \infty$. *Let* $\{\lambda_k\}$ *be the sequence of cosine coefficients of* f. *Then* $\sum_{k=1}^{\infty} \lambda_k$ *converges. Assume that* $\{\Lambda_n\} \in QD$. *Then for* $0 < q \leqslant \infty$,

$$\big\|\{\Lambda_n\}\big\|_{l(p',q)} \leqslant C \left\| \frac{f(x) - A}{x} \right\|_{L(p,q)(\mathbf{I})} \tag{23}$$

and for $1 < q < \infty$,

$$\big\|\{\Lambda_n\}\big\|_{l(p',q)} \leqslant C \left\| \frac{f(x) - A}{x} \right\|_{L^q_{\omega(p,q)}(\mathbf{I})}. \tag{24}$$

Proof. Let us see that $\sum_{k=1}^{\infty} \lambda_k$ converges in either case. Assume that $\{\lambda_n\}$ are the cosine coefficients of f. Let D_n be the Dirichlet kernel,

$$\begin{aligned}
\left| \sum_{k=m+1}^{n} \lambda_k \right| &= \left| \frac{2}{\pi} \int_0^{\pi} f(x)\big(D_n(x) - D_m(x)\big)\,dx \right| \\
&= \left| \frac{2}{\pi} \int_0^{\pi} \big(f(x) - A\big)\big(D_n(x) - D_m(x)\big)\,dx \right| \\
&= \left| \frac{2}{\pi} \int_0^{\pi} \big(f(x) - A\big) \frac{\sin(n+\frac{1}{2})x - \sin(m+\frac{1}{2})x}{2\sin\frac{x}{2}}\,dx \right|
\end{aligned}$$

and the last expression converges to 0 by the Riemann–Lebesgue Lemma. The proof for the sine coefficients is similar. Observe that for this part of the proof we need to assume only $\frac{f(x)-A}{x} \in L^1$ or $\frac{f(x)}{x} \in L^1$.

We proceed to prove (21). Of course $\frac{f(x)}{x} \in L(p,q)$ implies $f \in L(p,q)$. By Theorem 7, for $1 < p < \infty$, $0 < q \leqslant \infty$,

$$\big\|\{\Lambda_n\}\big\|_{l(p',q)} \leqslant C \left\| \sum_{n=1}^{\infty} \Lambda_n \cos nx \right\|_{L(p,q)(\mathbf{I})}.$$

By Lemma 4,

$$\begin{aligned}
&\left\| \sum_{n=1}^{\infty} \Lambda_n \cos nx \right\|_{L(p,q)(\mathbf{I})} \\
&\quad \leqslant C \left(\left\| \frac{\cos \frac{1}{2}x}{2 \sin \frac{1}{2}x} \sum_{n=1}^{\infty} \lambda_n \sin nx \right\|_{L(p,q)(\mathbf{I})} + \frac{1}{2} \left\| \sum_{n=1}^{\infty} \lambda_n (1 - \cos nx) \right\|_{L(p,q)(\mathbf{I})} \right) \\
&\quad \leqslant C \left(\left\| \frac{f(x)}{x} \right\|_{L(p,q)(\mathbf{I})} + \left\| \sum_{n=1}^{\infty} \lambda_n (1 - \cos nx) \right\|_{L(p,q)(\mathbf{I})} \right).
\end{aligned}$$

Since $f \in L(p,q)$, $Hf(x) = -\sum_{n=1}^{\infty} \lambda_n \cos nx \in L(p,q)$. By Theorem 8,

$$xH\left(\frac{f(y)}{y} \right)(x) = Hf(x) - Hf(0) = \sum_{n=1}^{\infty} \lambda_n (1 - \cos nx).$$

Therefore,

$$\left\| \sum_{n=1}^{\infty} \lambda_n (1 - \cos nx) \right\|_{L(p,q)(\mathbf{I})} = \left\| xH\left(\frac{f(y)}{y} \right)(x) \right\|_{L(p,q)(\mathbf{I})} \leqslant C(p) \left\| \frac{f(y)}{y} \right\|_{L(p,q)(\mathbf{I})}$$

and so

$$\left\| \{\Lambda_n\} \right\|_{l(p',q)} \leqslant C \left\| \sum_{n=1}^{\infty} \Lambda_n \cos nx \right\|_{L(p,q)(\mathbf{I})} \leqslant C \left\| \frac{f(x)}{x} \right\|_{L(p,q)(\mathbf{I})}.$$

Statement (22) is proved similarly by using the continuity of the Hilbert transform with respect to the $L^q_{\omega(p,q)}$ norm; see [6].

Assume that $f(x) = \sum_{n=1}^{\infty} \lambda_n \cos nx$. Since $f \in L(p,q)$, $Hf(x) = \sum_{n=1}^{\infty} \lambda_n \sin nx \in L(p,q)$ and $Hf(0) = 0$. By Theorem 8,

$$H\left(\frac{f(y) - A}{y} \right)(x) = \frac{Hf(x) - Hf(0)}{x} = \frac{1}{x} \sum_{n=1}^{\infty} \lambda_n \sin nx.$$

Therefore, by the first part of this theorem,

$$\begin{aligned}
\left\| \{\Lambda_n\} \right\|_{l(p',q)} &\leqslant C \left\| \frac{1}{x} \sum_{n=1}^{\infty} \lambda_n \sin nx \right\|_{L(p,q)(\mathbf{I})} = C \left\| H\left(\frac{f(y) - A}{y} \right) \right\|_{L(p,q)(\mathbf{I})} \\
&\leqslant C \left\| \frac{f(y) - A}{y} \right\|_{L(p,q)(\mathbf{I})}
\end{aligned}$$

proving (23). Statement (24) is proved similarly. □

Clearly,

Lemma 6. *Assume that f, defined on $(0,a)$, $0 < a \leqslant \infty$, has a nonincreasing rearrangement. Then for all $0 < b < a$,*

$$\left(f\left(|s - b| \right) \right)^*(x) \leqslant 2 f^*(x) I_{(0,\max\{b,a-b\})}(x).$$

Proof.

$$\begin{aligned}\big(f\big(|s-b|\big)\big)^*(x) &= \big(f(b-s)I_{(0,b)}(s) + f(s-b)I_{(b,a)}(s)\big)^*(x)\\ &= \big(f(t)I_{(0,b)}(b-t) + f(t)I_{(b,a)}(b+t)\big)^*(x)\\ &= \big(f(t)I_{(0,b)}(t) + f(t)I_{(0,a-b)}(t)\big)^*(x)\\ &\leqslant \big(2f(t)I_{(0,\max\{b,a-b\})}(t)\big)^*(x) \leqslant 2f^*(x)I_{(0,\max\{b,a-b\})}(x). \qquad \square\end{aligned}$$

Theorem 11. *Assume that* $\{\lambda_k\} \in l^1$, $0 \leqslant a < \pi$, $1 < p < \infty$, $0 < q \leqslant \infty$, *and* $g(x) = \sum_{k=1}^{\infty} \lambda_k \sin kx$ *or* $g(x) = \sum_{k=0}^{\infty} \lambda_k \cos kx$. *Then*

$$\left\| \frac{g(x)-g(a)}{x-a} \right\|_{L^q_{|x-a|^{\frac{1}{p}-\frac{1}{q}}}(\mathbf{I})} \leqslant C \left\| \left\{ \sum_{k=n}^{\infty} |\lambda_k| \right\} \right\|_{l(p',q)}. \tag{25}$$

Also,

$$\left\| \frac{g(x)-g(a)}{x-a} \right\|_{L(p,q)(\mathbf{I})} \leqslant C \left\| \left\{ \sum_{k=n}^{\infty} |\lambda_k| \right\} \right\|_{l(p',q)}. \tag{26}$$

Proof. In the sequel we agree that if $y < 1$, then $\sum_{k=1}^{y} \alpha_k = 0$. Let $g(x) = \sum_{k=1}^{\infty} \lambda_k \sin kx$ and $B_n = \sum_{k=n}^{\infty} |\lambda_k|$.

Inequality (25) is a part of Theorem 7 in [1]; we give an alternate proof, using interpolation techniques, that is both simpler and at the same time yields (26):

$$\left| \frac{g(x)-g(a)}{x-a} \right| \leqslant \sum_{k=1}^{\infty} |\lambda_k| \frac{|\sin kx - \sin ka|}{|x-a|}$$

so that

$$\left\| \frac{g(x)-g(a)}{x-a} \right\|_{L^\infty} \leqslant \sum_{k=1}^{\infty} k|\lambda_k| = \big\|\{\lambda_k\}\big\|_{l^1_k} \tag{27}$$

as well as

$$\left\| \frac{g(x)-g(a)}{x-a} \right\|_{L^\infty_{|x-a|}} \leqslant 2\sum_{k=1}^{\infty} |\lambda_k| = 2\big\|\{\lambda_k\}\big\|_{l^1}. \tag{28}$$

By Lemma 6,

$$\begin{aligned}\left\| \frac{g(x)-g(a)}{x-a} \right\|_{L(1,\infty)} &= \sup_{0<t<\pi} t\left(\frac{g(x)-g(a)}{x-a}\right)^*(t)\\ &\leqslant \big\|g(x)-g(a)\big\|_{L^\infty} \sup_{0<t<\pi} t\left(\frac{1}{x-a}\right)^*(t) \leqslant 4\big\|\{\lambda_k\}\big\|_{l^1}. \end{aligned} \tag{29}$$

Interpolating between (27) and (28),

$$\left\| \frac{g(x)-g(a)}{x-a} \right\|_{(L^\infty, L^\infty_{|x-a|})_{\frac{1}{p},q}} \leqslant C\big\|\{\lambda_k\}\big\|_{(l^1_k, l^1)_{\frac{1}{p},q}}.$$

Using Theorem 1, and later in the argument the continuity of g,

$$
\begin{aligned}
&C\left\|\frac{g(x)-g(a)}{x-a}\right\|_{(L^\infty,L^\infty_{|x-a|})_{\frac{1}{p},q}}\\
&\geqslant\left(\int_0^\infty\left(t^{-\frac{1}{p}}\left\|\frac{g(x)-g(a)}{x-a}I_{\{|x-a|>\frac{1}{t}\}}(x)\right\|_{L^\infty}\right)^q\frac{dt}{t}\right)^{\frac{1}{q}}\\
&=\left(\int_0^\infty t^{\frac{q}{p}-1}\left\|\frac{g(x)-g(a)}{x-a}\big(I_{(a+t,\pi)}(x)+I_{(0,a-t)}(x)\big)\right\|^q_{L^\infty}dt\right)^{\frac{1}{q}}\\
&\geqslant\max\left\{\left(\int_0^{\pi-a}t^{\frac{q}{p}-1}\left\|\frac{g(x)-g(a)}{x-a}I_{(a+t,\pi)}(x)\right\|^q_{L^\infty}dt\right)^{\frac{1}{q}},\right.\\
&\qquad\left.\left(\int_0^a t^{\frac{q}{p}-1}\left\|\frac{g(x)-g(a)}{x-a}I_{(0,a-t)}(x)\right\|^q_{L^\infty}dt\right)^{\frac{1}{q}}\right\}\\
&\geqslant\max\left\{\left(\int_0^{\pi-a}t^{\frac{q}{p}-1}\left|\frac{g(a+t)-g(a)}{t}\right|^q dt\right)^{\frac{1}{q}},\left(\int_0^a t^{\frac{q}{p}-1}\left|\frac{g(a-t)-g(a)}{-t}\right|^q dt\right)^{\frac{1}{q}}\right\}\\
&=\max\left\{\left(\int_a^\pi(x-a)^{\frac{q}{p}-1}\left|\frac{g(x)-g(a)}{x-a}\right|^q dx\right)^{\frac{1}{q}},\left(\int_0^a(a-x)^{\frac{q}{p}-1}\left|\frac{g(x)-g(a)}{x-a}\right|^q dx\right)^{\frac{1}{q}}\right\}\\
&\geqslant\left(\frac{1}{2}\int_0^\pi|x-a|^{\frac{q}{p}-1}\left|\frac{g(x)-g(a)}{x-a}\right|^q dx\right)^{\frac{1}{q}}.
\end{aligned}
$$

Again using Theorem 1,

$$
\begin{aligned}
\big\|\{\lambda_k\}\big\|_{(l^1_k,l^1)_{\frac{1}{p},q}}&=\big\|\{\lambda_k\}\big\|_{(l^1,l^1_k)_{\frac{1}{p'},q}}\\
&\sim\left(\int_0^\infty\left(t^{-\frac{1}{p'}}\left(\sum_{1\leqslant k<\frac{1}{t}}|\lambda_k|tk+\sum_{k\geqslant\frac{1}{t}}|\lambda_k|\right)\right)^q\frac{dt}{t}\right)^{\frac{1}{q}}\\
&\leqslant C\left(\left(\int_0^1\left(t^{\frac{1}{p}}\sum_{1\leqslant k<\frac{1}{t}}k|\lambda_k|\right)^q\frac{dt}{t}\right)^{\frac{1}{q}}+\left(\int_0^\infty\left(t^{-\frac{1}{p'}}\sum_{k\geqslant\frac{1}{t}}|\lambda_k|\right)^q\frac{dt}{t}\right)^{\frac{1}{q}}\right)\\
&=C\left(\left(\int_1^\infty\left(t^{-\frac{1}{p}}\sum_{1\leqslant k<t}k|\lambda_k|\right)^q\frac{dt}{t}\right)^{\frac{1}{q}}+\left(\int_0^\infty\left(t^{\frac{1}{p'}}\sum_{k\geqslant t}|\lambda_k|\right)^q\frac{dt}{t}\right)^{\frac{1}{q}}\right)\\
&=C\left(\left(\sum_{n=1}^\infty\int_n^{n+1}t^{-\frac{q}{p}-1}\left(\sum_{1\leqslant k<t}k|\lambda_k|\right)^q dt\right)^{\frac{1}{q}}\right.
\end{aligned}
$$

$$+\left(\sum_{n=1}^{\infty}\int_{n-1}^{n} t^{\frac{q}{p'}-1}\left(\sum_{k=n}^{\infty}|\lambda_k|\right)^q dt\right)^{\frac{1}{q}}\Bigg)$$

$$\leqslant C\left(\left(\sum_{n=1}^{\infty} n^{-\frac{q}{p}-1}\left(\sum_{k=1}^{n} k|\lambda_k|\right)^q\right)^{\frac{1}{q}}+\left(\sum_{n=1}^{\infty} n^{\frac{q}{p'}-1}\left(\sum_{k=n}^{\infty}|\lambda_k|\right)^q\right)^{\frac{1}{q}}\right)$$

$$=C\left(\left(\sum_{n=1}^{\infty} n^{\frac{q}{p'}-1}\left(\frac{1}{n}\sum_{k=1}^{n} k|\lambda_k|\right)^q\right)^{\frac{1}{q}}+\left\|\{B_k\}\right\|_{l(p',q)}\right).$$

Using Lemma 3, we shall see that

$$\left(\sum_{n=1}^{\infty} n^{\frac{q}{p'}-1}\left(\frac{1}{n}\sum_{k=1}^{n} k|\lambda_k|\right)^q\right)^{\frac{1}{q}}\leqslant C\left\|\{B_k\}\right\|_{l(p',q)}.$$

Let

$$G(y)=\sum_{k=1}^{\lfloor y\rfloor}|\lambda_k|.$$

Using (13),

$$\left(\sum_{n=1}^{\infty} n^{\frac{q}{p'}-1}\left(\frac{1}{n}\sum_{k=1}^{n} k|\lambda_k|\right)^q\right)^{\frac{1}{q}}$$

$$=\left(\sum_{n=1}^{\infty}\left(n^{-\frac{1}{p}-\frac{1}{q}}\int_{(0,n]} y\,dG(y)\right)^q\right)^{\frac{1}{q}}\sim\left(\int_0^{\infty}\left(x^{-\frac{1}{p}}\int_0^{x} y\,dG(y)\right)^q\frac{dx}{x}\right)^{\frac{1}{q}}$$

$$\leqslant C\left(\int_0^{\infty}\left(x^{\frac{1}{p'}}\int_x^{\infty} dG(y)\right)^q\frac{dx}{x}\right)^{\frac{1}{q}}\sim\left(\sum_{n=1}^{\infty}\left(n^{\frac{1}{p'}-\frac{1}{q}}\int_{[n,\infty)} dG(y)\right)^q\right)^{\frac{1}{q}}$$

$$=\left(\sum_{n=1}^{\infty} n^{\frac{q}{p'}-1}\left(\sum_{k=n}^{\infty}|\lambda_k|\right)^q\right)^{\frac{1}{q}}=\left\|\{B_k\}\right\|_{l(p',q)}.$$

This gives us (25). The proof for the cosine series is similar.

Interpolating between (27) and (29), we obtain

$$\left\|\frac{g(x)-g(a)}{x-a}\right\|_{(L^{\infty},L(1,\infty))_{\frac{1}{p},q}}\leqslant C\left\|\{\lambda_k\}\right\|_{(l_k^1,l^1)_{\frac{1}{p},q}}$$

that is to say,

$$\left\|\frac{g(x)-g(a)}{x-a}\right\|_{L(p,q)}\leqslant C\left\|\{B_k\}\right\|_{l(p',q)}$$

proving (26). Again, the proof for the cosine series is similar. □

Theorem 12. *Assume that* $\sum_{k=1}^{\infty}\lambda_k$ *converges and* $\{\Lambda_n\}\in QD\cap l(p',q)$, *where* $1<p<\infty$ *and* $0<q\leqslant\infty$. *Then* $\sum_{n=1}^{\infty}\lambda_n\sin nx$ *and* $\sum_{n=1}^{\infty}\lambda_n\cos nx$ *converge. For all* $0\leqslant a<\pi$, *and for* $g(x)=\sum_{n=1}^{\infty}\lambda_n\sin nx$ *or* $g(x)=\sum_{n=1}^{\infty}\lambda_n\cos nx$, *we have*

$$\left\|\frac{g(x)-g(a)}{x-a}\right\|_{L(p,q)(\mathbf{I})}\leqslant C(p)\big\|\{\Lambda_n\}\big\|_{l(p',q)}. \tag{30}$$

Also,

$$\left\|\frac{g(x)-g(a)}{x-a}\right\|_{L^q_{|x-a|^{\frac{1}{p}-\frac{1}{q}}}(\mathbf{I})}\leqslant C(p)\big\|\{\Lambda_n\}\big\|_{l(p',q)}. \tag{31}$$

Proof. Let us assume first that $\Lambda_n\searrow 0$. Then $\{\lambda_n\}\in l^1$, and we obtain (30) by an application of Theorem 11.

If $\{\Lambda_n\}\in QD$, then by Corollary 2, there exist sequences $\{\Lambda_{j,n}\}$ so that $\Lambda_{j,n}\searrow 0$, $\Lambda_n=\Lambda_{2,n}-\Lambda_{1,n}$, and $\|\{\Lambda_n\}\|_{l(p',q)}\sim\|\{\Lambda_{j,n}\}\|_{l(p',q)}$. Let $g(x)=\sum_{n=1}^{\infty}\lambda_n\sin nx$ and $g_j(x)=\sum_{n=1}^{\infty}(\Lambda_{j,n}-\Lambda_{j,n+1})\sin nx$. Then

$$\begin{aligned}\left\|\frac{g(x)-g(a)}{x-a}\right\|_{L(p,q)(\mathbf{I})}&=\left\|\frac{g_2(x)-g_2(a)}{x-a}-\frac{g_1(x)-g_1(a)}{x-a}\right\|_{L(p,q)(\mathbf{I})}\\&\leqslant C\left(\left\|\frac{g_2(x)-g_2(a)}{x-a}\right\|_{L(p,q)(\mathbf{I})}+\left\|\frac{g_1(x)-g_1(a)}{x-a}\right\|_{L(p,q)(\mathbf{I})}\right)\\&\leqslant C\big(\big\|\{\Lambda_{2,n}\}\big\|_{l(p',q)}+\big\|\{\Lambda_{1,n}\}\big\|_{l(p',q)}\big)\leqslant C\big\|\{\Lambda_n\}\big\|_{l(p',q)}.\end{aligned}$$

The same argument proves (30) for cosine series; (31) is proved in the same way. □

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